



AnalystLab Africa - HealthConnect Experience Lab

Data Science Feedback to Data Analytics - Week 6

In response to the Data Analytics track's Week 6 document ("Data Analytics Collaboration with Data Science Track"), the following feedback is provided as requested in Section 6 of that document.

A. Findings Used

All 5 findings were reviewed. Findings 1 (previous no-shows), 2 (booking lead time), 3 (reminder status), and 4 (distance) correspond to variables already present in the Data Science feature set since Week 5. Finding 5 (previous no-show as a high-risk segment flag) overlaps with an existing engineered feature (`high_risk_history`).

B. Modelling Impact

Yes two specific suggestions were tested and adopted:

1. A categorical lead-time band feature, matching the 0-7 / 8-14 / 15-30 / 31-60 day bands used in the Data Analytics analysis, tested alongside the existing continuous `booking_lead_days` variable.
2. A `previous_no_shows` × `booking_lead_days` interaction term.

Both were retained after measurably improving model performance (ROC-AUC increased from approximately 0.680 to 0.682 when both were added together). This directly influenced feature engineering and the final Week 6 candidate model.

C. Model Validation

Partially aligned, with one discrepancy worth highlighting. The Data Science model's feature importance ranking places `booking_lead_days` and `distance_to_clinic_km` above `previous_no_shows` in predictive importance even though the Data Analytics descriptive analysis shows a very strong, clear association for previous no-shows (43.5% to 68.8% no-show rate across the 0/1/2/3+ bands).

This suggests previous no-show history is a strong descriptive signal, but its incremental predictive contribution is partially absorbed by other correlated variables once all variables are considered together in a multivariate model. This is a useful nuance: a finding can be strongly associated with the outcome in isolation (Data Analytics' view) while contributing comparatively less once combined with other predictors (Data Science's view) both conclusions are valid and reflect different, complementary analytical lenses.

All four quantitative findings (previous no-shows, lead-time bands, reminder status, distance) were independently re-computed on the Data Science team's own prepared dataset and replicated closely (within a few percentage points), confirming both tracks are observing the same underlying patterns in the shared HealthConnect data.

D. Additional Analytical Requirements

It would be valuable for Data Analytics to check whether the 31-60 day lead-time segment's high no-show rate (60.5%) holds consistently across appointment types, or is concentrated in specific types. The Data Science error analysis found lead-time effects but did not cross-tabulate this with appointment type. Data Analytics' segment-level tools (e.g. the Power BI dashboard) may be better suited to that specific breakdown.

Summary of Validated Figures (Data Analytics vs. Data Science Recomputation)

Previous no-shows (0 / 1 / 2 / 3+): DA: 43.5% / 53.5% / 59.4% / 68.8% - DS recomputation: 46.3% / 55.9% / 62.1% / 70.3%

Lead time 31-60 days: DA: ~60.5% - DS recomputation: 63.9%

Reminder sent vs not sent: DA: 47.4% vs 51.4% - DS recomputation: 49.9% vs 54.6%

Distance $\geq$ 20km: DA: ~58% - DS recomputation: 60.2%

Thank you for the detailed and well-structured findings this was very useful input for Week 6 model improvement and validation.